Forecasting the Number of International Tourist Arrivals to Indonesia via Ngurah Rai Airport, Bali

Danella Citra Aldarena[1], Aulia Rahmaeta Octaviani[1], Widyana Nityasa Manda[1], Najwa Allena Eka Putri Santoso[1], Najwa Putri Lathifah[1]

[1] Department of Statistics, Institut Teknologi Sepuluh Nopember, Kampus ITS Sukolilo, Surabaya 60111, Indonesia

Abstract. The tourism sector is a major contributor to Indonesia's economy, with Bali Province as a leading destination. Fluctuations in foreign tourist arrivals through I Gusti Ngurah Rai International Airport, characterized by seasonal patterns, trends, and structural disruptions due to the COVID-19 pandemic, create complexity in forecasting modeling. This study aims to test and compare the accuracy of machine learning (Random Forest, LightGBM) and deep learning (Neural Hierarchical Interpolation for Time Series/N-HiTS) models in predicting the time series of foreign tourist arrivals. The evaluation method uses the Root Mean Squared Error (RMSE) and Symmetric Mean Absolute Percentage Error (sMAPE) metrics. The analysis results show that in the training data, Random Forest shows relatively stable performance. However, in the test data, the N-HiTS model significantly outperforms with the lowest RMSE and sMAPE values, indicating better generalization capacity in dealing with post-pandemic data structure changes. Based on these findings, N-HiTS was selected as the optimal model for future projections. The forecasting results revealed a short-term downward trend after the peak period, reflecting a consolidation phase in tourism recovery. The implications of this study highlight the effectiveness of multiresolution decomposition-based deep learning architecture for complex time series data affected by external interventions, which can serve as an analytical basis for strategic planning in the recovery of the tourism sector and optimization of airport capacity in Bali.

Keywords: Tourist Arrivals · Bali Tourism · Random Forest · LightGBM · N-HiTS Model · COVID-19 Impact

1 Introduction

Tourism is an economic sector that plays a significant role in driving global economic growth because it can generate foreign exchange, employment, and multiplier effects on other sectors. Before the COVID-19 pandemic, international travel showed a consistent upward trend, particularly in the Asia-Pacific region, which is one of the centers of global tourism growth [1]. Therefore, this study emphasizes the use of data-based and machine learning methods to improve the accuracy of tourism demand forecasting [2]. At the national level, Indonesia views tourism as a strategic sector that plays a significant role in strengthening the national economy, increasing foreign exchange earnings, and promoting national development. One example is the island of Bali, which is a leading destination and one of the main icons of Indonesian tourism, consistently contributing the largest number of foreign tourist visits [3]. The main access point used by foreign tourists is through I Gusti

Ngurah Rai International Airport, which is also one of the busiest airports in Indonesia. According to research [4], the number of foreign tourists arriving through this airport has a fluctuating pattern that can be influenced by seasonal factors, global demand dynamics, international travel policies, and external factors such as the COVID-19 pandemic.

The COVID-19 pandemic caused a significant decline in the number of foreign tourists in 2020, followed by a gradual recovery period from 2022 to 2024, which also coincided with the easing of global travel restrictions [3]. This situation has resulted in complex and non-linear tourist arrival data, characterized by long-term trends, strong seasonal fluctuations, and significant structural disturbances. Previous studies discussing tourist arrival forecasting in Indonesia, particularly in Bali, have been dominated by statistical methods such as SARIMA, Holt-Winters, and simple hybrid models. Although these methods capture regular seasonal patterns, their performance tends to decline when data experiences major disruptions and non-linear pattern changes, such as during the pandemic period [5] [6]. Therefore, recent research has begun to adopt machine learning and deep learning methods that are more flexible in modelling nonlinear relationships and complex interactions between variables.

With machine learning and deep learning approaches increasingly used in tourism demand forecasting because tourist arrival data tends to contain non-linear patterns, seasonality, and structural changes due to external shocks that are difficult to capture stably by a single model based on classical assumptions [7]. Based on the literature, it is also known that machine learning-based models can work well for predicting tourist visits and arrivals when explanatory variables evolve and when patterns change in the post-pandemic period, including applications in the context of Bali and daily visitor volumes that require pattern flexibility [8]. Deep learning, long-horizon time series models such as N-HiTS, are relevant for contexts with strong seasonality and regime shifts due to their focus on multi-scale forecasting for long horizons [9].

Although machine learning approaches have been widely applied in time series forecasting and show promising results, studies that directly compare modern ensemble methods and time series-specific deep learning architectures for international tourist arrivals through I Gusti Ngurah Rai International Airport are still limited. Previous research has tended to evaluate one approach separately without systematic comparison, particularly in the context of [4]. This indicates the need for a study that assesses the relative performance of various forecasting approaches to determine the most appropriate method for handling fluctuating tourist arrival data.

Based on the background and focus of the problem described above, this study aims to develop and compare models for forecasting the number of foreign tourist arrivals to Indonesia through I Gusti Ngurah Rai International Airport, Bali, using the Random Forest, LightGBM, and N-HiTS methods. Through this comparison, this study is expected to contribute to the selection of the appropriate forecasting approach and serve as a basis for data-driven decision making to support tourism recovery planning and air transportation management in Bali.

2 Literature Review

2.1 Statistical Machine Learning

Statistical machine learning represents a transformative framework that merges classical statistical theory with computational algorithms to prioritize generalization over simple data fitting. As established by James et al. [10], the core challenge of any learner is the "Bias-Variance Tradeoff," where a model must be flexible enough to capture signals but rigid enough to ignore noise. Furthermore, Hastie et al. [11] framework treats machine learning as an exercise in "Empirical Risk Minimization," where the goal is to find a function f(x) that minimizes the expected loss across unseen observations. These

foundational texts argue that the validity of a model is not just in its predictive accuracy, but in its ability to be validated through rigorous resampling methods like Cross-Validation.

2.2 N-HiTS Method

N-HiTS is a neural network–based time series forecasting model designed to improve long-term prediction accuracy. It works by using hierarchical interpolation and multi-rate sampling to capture both long-term trends and short-term seasonality, making it particularly suited for data with abrupt disruptions and seasonal volatility. These techniques help the model produce stable and reliable forecasts across different time scales, while also reducing computational cost. Research results show that N-HiTS performs better than several modern forecasting methods, especially for long-horizon predictions [9].

N-HiTS also has the ability to break down data patterns into simpler components, making the forecasting process easier to interpret. This allows the model to capture important structures in time series data more clearly, without relying on overly complex architectures. N-Hits also requires less computation time and memory compared to similar models. Based on tests by Challu et al. [9], N-HiTS achieved lower prediction errors and ran more efficiently than other similar models. The results also showed that it used less computation time and memory across multiple datasets, making it a strong choice for both research and real-world forecasting needs.

2.3 LightGBM Method

LightGBM is a gradient boosting framework that uses decision tree algorithms. It is a more efficient implementation of gradient boosting decision tree which is an ensemble model of decision trees trained in sequence and a widely-used machine learning algorithm due to its efficiency, accuracy, and interpretability [12].

A GBM improves model performance by focusing on errors from previous predictions and updating them step by step, unlike other methods such as random forest or bagging that build multiple trees independently. In LightGBM, the trees grow vertically using the leaf-wise strategy, where the model repeatedly splits the leaf with the highest loss value. This produces deeper and asymmetric tree that allows the model to reduce prediction error more effectively. The biggest advantage of leaf-wise tree growth is that it is fast and require less memory for execution, making LightGBM suited for handling large amounts of data. However, LightGBM is not recommended for small data sets because it is sensitive to overfitting and can easily overfit small data .

2.4 Random Forest Method

Random Forest is a widely used machine learning method known for producing highly accurate and stable predictions. According to the book "*An Introduction to Statistical Learning"*, this method works by combining many individual "decision trees" into one unified model. A major issue with using just one decision tree is that it can be unstable; small changes in the data can drastically change the final prediction. To fix this, Random Forest uses a technique called "bagging" (bootstrap aggregating), where the algorithm trains many different trees independently and then averages their results. This effectively lowers the risk of errors and makes the model much more consistent when dealing with different types of data.

One of the unique strengths of Random Forest, as explained by James et al. , is the use of "feature randomness". At every step of building a tree, the algorithm only looks at a small, random subset of variables. This ensures that the trees in the "forest" are different from one another, preventing the model from relying too heavily on a single dominant variable. Furthermore, Hastie et al. emphasize that this method is very practical because it can show which variables are the most important for a prediction. With an internal testing

feature called "Out-of-Bag" (OOB) error, researchers can validate the model's accuracy without needing complicated additional testing procedures.

2.5 Metric Evaluation

Assessing the accuracy of point forecasts is crucial for understanding how effectively a model predicts future values. The difference between the observed data and the forecasted results forms the basis for calculating prediction errors. These errors are then evaluated using several performance metrics that indicate how closely the forecasts align with the real data. Through these accuracy measures, it becomes possible to objectively compare forecasting models and gain clearer insight into their predictive capability.

RMSE

RMSE represents the average level of error between predicted values and actual observations. In other words, it reflects the typical vertical distance between real data points and their corresponding forecast values along a fitted line. Mathematically, RMSE is obtained by taking the square root of the mean squared error (MSE), making it a straightforward measure of overall prediction accuracy.

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}$$

sMAPE

sMAPE is a percentage-based accuracy metric considered an improved and more balanced form of MAPE. It applies equal weighting to positive and negative errors. Despite this advantage, sMAPE also comes with limitations. When actual values are close to zero, the metric can produce extremely large or unstable results, making interpretation more difficult. Additionally, sMAPE is more computationally complex than many other standard forecasting metrics.

$$sMAPE = \frac{100}{n}\sum_{t=1}^{n}\frac{|y_t - \hat{y}_t|}{(|y_t| + |\hat{y}_t|)/2}$$

2.6 Forecasting

Forecasting is the process of predicting future observations based on historical data and the structural patterns within a time series. Forecasts are generated by extrapolating components such as trend, seasonality, and irregular variations using estimated model parameters, allowing results to reflect both expected future values and the uncertainty linked to each element. Makridakis et al. note that forecasting relies on the assumption that past patterns will continue, making quantitative methods effective when sufficient numerical data and stable time structures are present. These approaches are widely used because they are practical, cost-efficient, and able to produce reliable short-term predictions that support planning and decision-making.

3 Dataset and Methodology

3.1 Research Data

This research is conducted through the following systematic stages:

1. Data Collection
   Secondary data on international tourist arrivals to Indonesia is collected from the Central Bureau of Statistics (BPS), covering the period from January 2010 to

December 2024. The dataset includes monthly records used for time series forecasting.

2. Preprocessing Data
   Data preprocessing involves:
   a. Cleaning and converting Indonesian month names to datetime format
   b. Feature engineering, including: COVID-19 dummy variable (March 2020–June 2022), Monthly seasonal dummy variables via one-hot encoding.
   c. Normalization using Min-Max Scaler to transform data into the range [0, 1]
3. Transformation Data
   Processed data is transformed into a structured format suitable for machine learning models :
   a. Conversion of the time series into supervised learning format using 12-period lags
   b. Splitting into training data (January 2010–December 2023) and testing data (January–December 2024)
   c. Application of Darts' TimeSeries format for model compatibility
4. Training Data
   The training dataset is used to build and train three forecasting models : N-Hits, LightGBM, and Random Forest.
5. Model Training
   Each model is trained with specific configurations and covariates :
   a. Random Forest and LightGBM are trained using future covariates (seasonal and COVID dummies)
   b. N-HiTS is trained using past covariates over the full historical period
   c. Hyperparameters are tuned to optimize performance (e.g., epochs, learning rate, tree depth)
6. Model Evaluation
   Model performance is evaluated using :
   a. RMSE (Root Mean Square Error) and MAPE (Mean Absolute Percentage Error)
   b. Visualization of predicted vs. actual values for the test period (2024)
   c. Comparison of results to select the best-performing model for forecasting

   The following is a flow chart of the research :

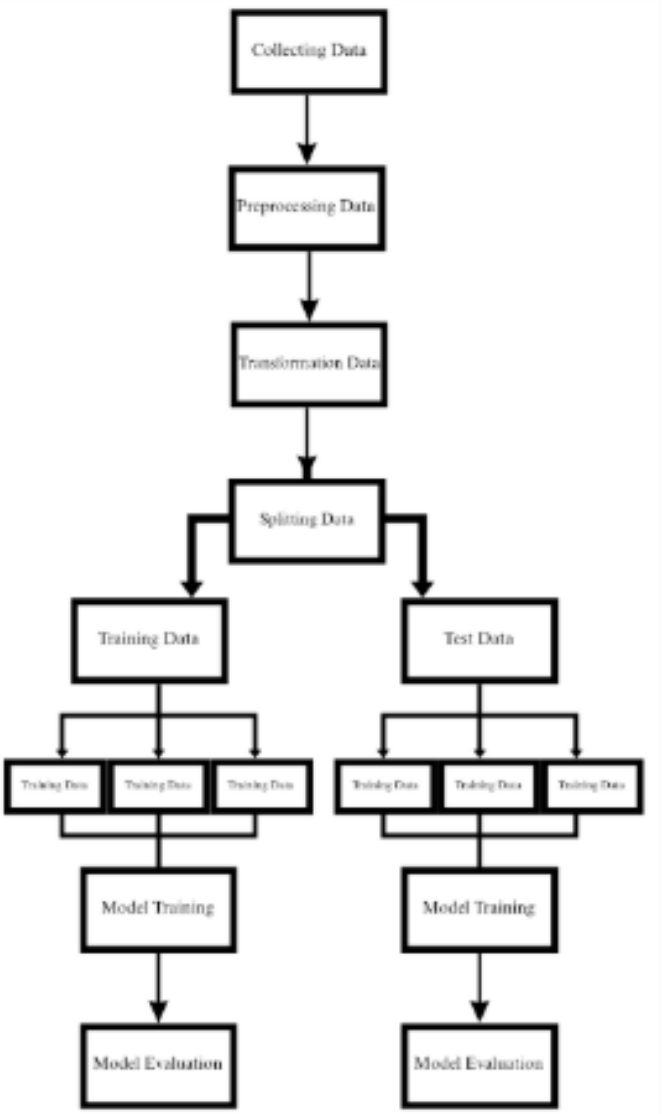


Fig. 1. Research Flow

4 Results and Evaluation

4.1 Description of Data and Time Series Patterns

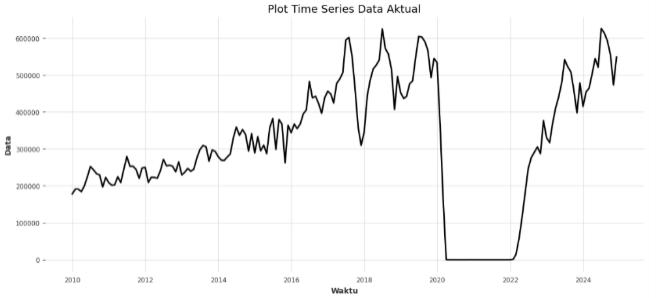


Fig. 2. Time Series Patterns

Based on the time series plot for the period 2010 to 2024, it can be seen that tourist arrival data shows a long-term upward trend with increasing fluctuations over time. In addition to the trend, the recurring annual ups and downs indicate a monthly seasonal component. From 2020 to mid-2022, there was an extreme decline to near zero, reflecting structural changes due to external events that cannot be explained by normal historical dynamics. After this period, the data shows a strong recovery phase, with values increasing significantly and, in the period 2023–2024, approaching and even exceeding pre-extreme decline levels. These characteristics indicate that the data contains trends, seasonality, and external shocks, requiring a forecasting approach capable of capturing complex dynamics.

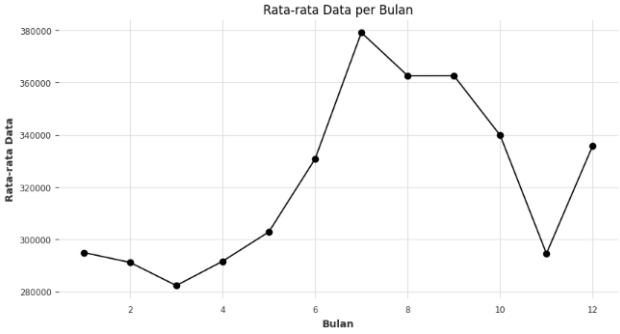


Fig. 3. Average Monthly Data

To further understand the seasonal patterns identified in the time series plot, the analysis continued by observing the behavior of the data based on monthly averages. Analysis of the monthly average data reinforced the existence of seasonal patterns. The average values tended to be lower at the beginning of the year and reached their lowest point in March, then gradually increased until they peaked in July. After that, the values decline again from August to October and experience a sharp decline in November, before increasing again in December. This pattern indicates a peak season in the middle of the year and a low season at the beginning and end of the year, so seasonal components need to be taken into account in the forecasting process.

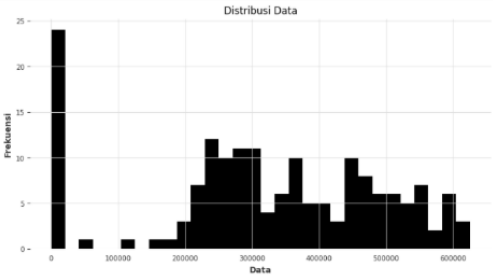


Fig. 4. Data Distribution

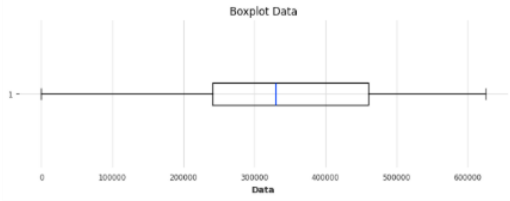


Fig. 5. Boxplot Data

In addition to trend and seasonal patterns, the characteristics of data distribution must also be considered to understand the level of variation and the existence of extreme values. The results of distribution analysis and boxplots show that the data has an asymmetrical distribution with a fairly wide range of values. Most observations are concentrated in the

middle to high range, but there are very small and very large values that indicate the existence of outliers. The median position around the middle of the data range indicates the dominance of middle values, while the long whiskers and extreme points confirm the existence of data spikes in certain periods, which are likely related to specific events or external interventions. These findings indicate that the data is volatile and unstable, making it relevant to select a forecasting method that is robust against variation and outliers.

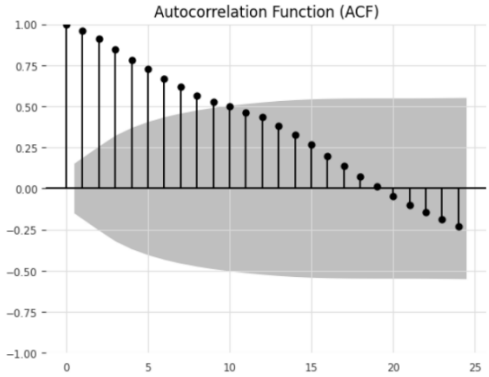


Fig. 6. ACF Plot

Next, the ACF plot of the data is analyzed. The autocorrelation function (ACF) is used to identify temporal dependencies in the time series. The ACF plot shows strong positive autocorrelation at the initial lags, with correlation values gradually decreasing as the lag increases. This slow decay pattern indicates the presence of a trend component and strong time dependence in the data. At higher lags, the influence of past observations weakens. No clear or consistent seasonal spike pattern is observed in the ACF plot, suggesting that seasonal behavior cannot be conclusively identified from the ACF alone and should be confirmed through additional analyses.

4.2 Splitting Data and Evaluation Scheme

At this stage, the data is divided into two parts, namely train data and test data, taking into account the chronological nature of time series. The division is based on time sequence, where train data covers all observations up to December 2023, while test data starts from January 2024 until the end of the observation period. This chronological data division approach is applied to maintain the principle of time-aware validation, which ensures that information from the future is not used in the model training process. Thus, the model is trained using only historical data available at the time of prediction, so that the evaluation of model performance on test data reflects more realistic forecasting conditions. The training data is used to build and adjust model parameters, while the test data is used to evaluate the model's ability to make predictions in periods not involved in the training process. This scheme allows for a more objective assessment of model performance, especially in the context of tourist arrival data, which has undergone significant changes in patterns and dynamics in recent periods.

4.3 Model Performance Comparison on Training Data

Model performance comparison is conducted by observing prediction results on training data and test data simultaneously, both through evaluation metrics and visual patterns between actual values and predicted values. This approach is used to obtain a more comprehensive picture of each model's ability to capture historical patterns, respond to structural changes, and generalize over previously unseen periods.

At this stage, model training was conducted using training data up to December 2023 to evaluate the initial capabilities of each method in capturing historical patterns of tourist arrival data. Performance evaluation on the training data was carried out using two metrics,

namely Root Mean Squared Error (RMSE) and symmetric Mean Absolute Percentage Error (sMAPE), which represent the absolute error and relative error of the model, respectively.

Table 1. Model Performance Comparison on Training Data

	RMSE	sMAPE
Random Forest	15018.750511	25.358148
Light GBM	25222.335100	36.866023
NHiTS	15021.294633	37.466178

Based on Table 1 it is known that the evaluation results show that the Random Forest and N-HiTS models have relatively similar RMSE values in the training data, 15018.75 and 15021.29. These values are lower than LightGBM, which produced an RMSE of 25222.34. This indicates that Random Forest and N-HiTS are better at following historical data patterns in the training phase than LightGBM. In terms of sMAPE, Random Forest shows more stable performance with a relative error value of 25.36 percent. Meanwhile, LightGBM and N-HiTS have higher sMAPE values, at 36.87 percent and 37.47 percent, respectively. This difference shows that even though N-HiTS has the ability to capture nonlinear and complex patterns, the relative deviation from the actual values in the training data is still quite large.

Overall, the model training results show that Random Forest has a better balance between absolute error and relative error in the training data. However, performance on the training data cannot be used as the sole basis for determining the best model, because the model's generalization ability to the test data remains a major factor in the final evaluation.

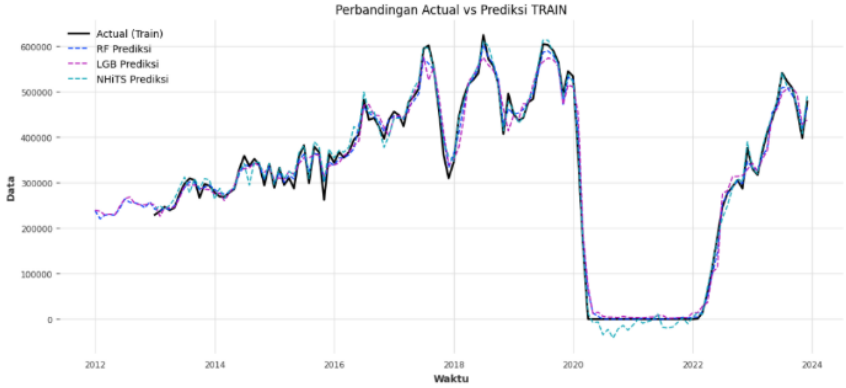


Fig. 7. Model Performance Comparison Plot on Training Data

In the training data, the comparison plot between actual and predicted values shows that all three models, namely Random Forest, LightGBM, and NHiTS, are generally able to follow the upward trend and seasonal fluctuations found in the data. All models successfully captured the long-term upward pattern and the drastic decline that occurred during the extreme intervention period around 2020. However, there are differences in the responses between models. Random Forest and LightGBM tend to follow pattern changes more smoothly, while NHiTS shows a sharper response when extreme declines occur, enabling it to represent structural changes more clearly. In the recovery phase after the intervention period ended, all three models were again able to follow the upward pattern of the actual data quite well.

4.4 Model Performance Comparison on Testing Data

After evaluating the model on the training data, the next critical step is to test the robustness of the model on (testing data. This section analyzes the performance of Random Forest, LightGBM, and NHiTS in predicting test data to determine the model with the best generalization ability. The complete evaluation results are presented in Table 2 below.

Table 2. Model Performance Comparison on Testing Data

	RMSE	sMAPE
Random Forest	38990.452953	6.487435
Light GBM	41624.668209	6.677481
NHiTS	35581.002546	5.065780

In the test data, the differences in performance between the models are more apparent. All three models were able to capture the direction of the actual data throughout 2024, but the degree of closeness of the predictions to the actual values differed. The NHiTS model visually showed the closest predictions to the actual data, especially in following the rise to the peak in the middle of the year. Although in some months NHiTS tended to provide estimates that were slightly higher than the actual values, this model was still able to respond quickly to changes in data levels. Random Forest produced more stable predictions and tended to be below the actual values, especially during periods of rapid increase, indicating a tendency to underestimate. LightGBM was between the two models, with a relatively consistent prediction pattern but less responsive to sharp changes in levels.

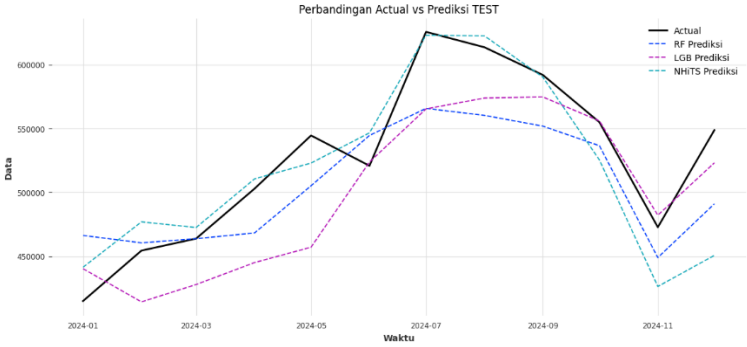


Fig. 8. Model Performance Comparison Plot on Testing Data

This comparison is consistent with the results of quantitative evaluation of the test data, where NHiTS produced the lowest RMSE and sMAPE values compared to other models. These findings indicate that NHiTS has better generalization capabilities in dealing with changes in patterns after extreme interventions. Random Forest continues to show competitive performance with relatively stable predictions, while LightGBM shows limitations in capturing significant structural changes. Overall, the comparison of model performance shows trade-offs between approaches. NHiTS excels at capturing complex data patterns and dynamics, especially in the recovery phase, making it most suitable for advanced forecasting needs. Random Forest offers prediction stability with lower sensitivity to extreme fluctuations, while LightGBM shows more moderate and conservative performance in responding to data pattern changes.

5 Discussion and Implication

5.1 Discussion of Model Comparison Results

The differences in performance between models in Chapter 4 reflect the different characteristics of each approach in responding to tourist arrival data that experiences pattern changes and extreme interventions. The data analyzed not only contains trends and seasonality, but also structural changes, thus requiring models that are able to adapt to these dynamics. Ensemble tree-based models tend to produce more stable predictions due to aggregation mechanisms that dampen fluctuations, but are less responsive to rapid changes in levels. Conversely, deep learning approaches have a higher adaptability to pattern

changes, making them more suitable for data with rapid recovery phases. These differences in character indicate that model selection needs to consider the balance between stability and adaptability, not just accuracy.

5.2 Model Selection for Forecasting

Based on the results of quantitative evaluation of test data and visual patterns of prediction results, the NHiTS model was selected as the main model for forecasting in the coming period. This selection is based on the performance of NHiTS, which consistently shows the lowest error rate compared to other models, both in terms of absolute and relative error, thus reflecting better generalization capabilities on previously unobserved data. In addition to quantitative performance, NHiTS also shows better adaptability to structural changes in the data. The structure of tourist arrival data, which experienced an extreme decline during the COVID-19 pandemic period followed by a rapid recovery phase, requires a model capable of capturing sharp changes in patterns. The analysis results show that NHiTS is more responsive in following these dynamics compared to Random Forest and LightGBM, which tend to produce smoother and more conservative predictions.

The difference in performance between the training and testing phases was also taken into consideration in the model selection. Although the sMAPE value of NHiTS on the training data was relatively higher than some other models, its performance actually improved on the test data. This indicates that NHiTS did not experience overfitting and had good generalization capabilities, making it more reliable for use in medium-term forecasting. For future forecasting purposes, the NHiTS model was then retrained using all available historical data up to December 2024. The purpose of this retraining process is to enable the model to learn more complete data patterns, both before and after extreme interventions, so that the resulting projections are more stable and representative. With this approach, forecasts are made for the next 12-month horizon, namely the period from January 2025 to December 2025, in order to provide an overview of tourist arrivals in the coming period and support data-driven decision making.

5.3 Tourist Arrivals Projection

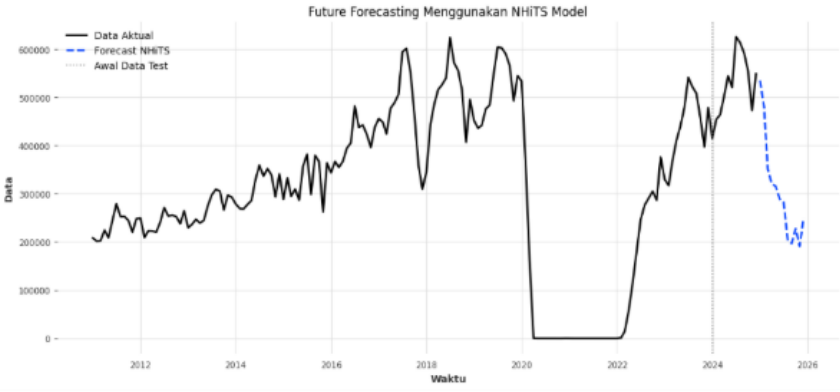


Fig. 9. Tourist Arrivals Projection

The results of forecasting using the NHiTS model provide an overview of the development of foreign tourist arrivals in the coming period after the last historical data. Based on the projection results, the model predicts a decline in value at the beginning of the forecasting period, followed by fluctuations in data at a lower level compared to historical values prior to the projection period. This pattern indicates that although the recovery trend is maintained, the growth rate in the early forecasting period tends to be more moderate.

The fluctuations generated in the projection period have a relatively smaller amplitude compared to the fluctuations in historical data. This indicates that the model projects more

stable seasonal variations without the extreme spikes or declines that occurred during the pandemic period. This characteristic shows that the model assumes that data conditions have returned to a more normal and stable pattern, so it does not include scenarios of major disruptions in short-term projections. The decline seen at the beginning of the forecasting period can be interpreted as the model's adjustment response after the last observation period, where historical values were at a relatively high level. After this adjustment phase, the projection values move more smoothly and follow long-term historical patterns, reflecting NHiTS's ability to produce smooth and stable forecasts.

Overall, the projection results show that the NHiTS model is capable of providing a moderate and realistic picture of tourist arrivals in the coming period. These projections can be used as a basis for medium-term planning, with the caveat that the forecast results do not assume any sudden major external interventions.

5.4 Implications of Research Results

The results of forecasts using the NHiTS model show that the arrival of foreign tourists to Bali in the coming period is expected to be more stable without extreme fluctuations such as those that occurred during the pandemic period. This moderate projection pattern indicates that the tourism recovery phase is proceeding gradually and in a relatively controlled manner.

For the management of I Gusti Ngurah Rai International Airport, these forecast results can be used as a basis for operational planning, particularly in relation to service capacity adjustments, human resource management, and the readiness of supporting facilities. Information on tourist movement patterns enables more measured decision-making and reduces uncertainty in daily operational management. In addition, for local governments and tourism industry players, the findings of this study can be used as a reference in formulating more targeted tourism recovery and development strategies. The data-based approach used in this study is expected to support the formulation of more adaptive policies and increase the effectiveness of tourism planning in facing future demand dynamics.

6 Conclusion

This study aims to analyze and model foreign tourist arrival data, which has complex characteristics such as long-term trends, seasonal patterns, and extreme interventions due to the COVID-19 pandemic. Exploratory analysis results show that the data is unstable, contains outliers, and has undergone significant structural changes, requiring a forecasting approach that can capture nonlinear dynamics and pattern changes adaptively. A comparison of the performance of the Random Forest, LightGBM, and NHiTS models shows differences in their characteristics in responding to the data. In the training data, Random Forest showed a good balance between absolute and relative errors, while NHiTS and LightGBM had larger relative errors. However, evaluation on the test data showed that NHiTS performed best with the lowest RMSE and sMAPE values, indicating better generalization ability to previously unobserved data.

Based on the quantitative evaluation results and visual analysis, the NHiTS model was selected as the best model for forecasting. The advantage of NHiTS lies in its ability to capture complex patterns, respond to changes in data structure, and maintain prediction stability after periods of extreme intervention. This model has been proven to not experience overfitting and is able to adapt to the post-pandemic recovery phase. The results of projections using the NHiTS model show that the arrival of foreign tourists in the coming period is expected to be more stable with more moderate fluctuations compared to the extreme historical period. These projections provide a realistic picture of future tourism conditions, assuming there are no major external disruptions. Thus, the results of this study can serve as a supporting basis for strategic planning and data-driven decision-making in

the tourism and air transportation sectors.

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